1 Lecture 17: Plan for Class

Last time:

1. Smale horseshoe where the initial set is
   (a) a Cantor set
   (b) measure zero
   (c) uncountable ∞
2. The dynamics of the Smale horseshoe are the same as a shift on two symbols

Today:

1. Chain recurrence §5.2
2. Hyperbolic structure §5.2
3. Stable manifold theorem for hyperbolic sets §5.2
4. Λ lemma §5.2
5. Markov partitions §5.3
6. Subshift of finite type §5.3
7. Sketch proof of Smale-Birkhoff theorem §5.3

2 Recurrence

We have a few different objects in recurrence

- fixed point
  - You approach some point asymptotically
- periodic orbit
  - You follow some cycle upon further applications of recurrence
- nonwandering point
  - If you start at some point, you will return to a neighborhood of that point some amount of time later
- Chain recurrence

2.1 Chain recurrence

<u>Definition 5.2.3</u> A point x is chain recurrent for φ_t if $\forall \varepsilon > 0$, $\exists x = x_0, x_1, x_2, x_3, \ldots, x_n = x$ and times $t_1, t_2, \ldots, t_n$ such that

$$\|\|\varphi_{t_i}(x_{i-1}) - x_i\|\| < \varepsilon, \quad i = 1, \ldots, n$$

This is just approximating a periodic orbit where you assume some error ε. Note that $x = x_0$ and $x_n = x$. This is just the starting and end points are kind of the same

3 Hyperbolic structure

<u>Definition 5.2.6</u> A hyperbolic structure for an invariant set Λ of $F : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous[1], invariant direct sum decomposition

$$T_\Lambda \mathbb{R}^n = E^s_\Lambda \oplus E^u_\Lambda$$

with the properties that

- $\exists c > 0$, $0 < \lambda < 1$ such that
- $v \in E^u_x \implies ||Df^{-n}(x)v|| < c\lambda^n \, ||v||$
- $v \in E^s_x \implies ||Df^{n}(x)v|| < c\lambda^n \, ||v||$
- for all $n = 1, 2, \ldots$

This just says that points in the unstable direction get squished by a factor of λ^n when iterating backward n times, and that points in the stable direction get squished by a factor of λ^n when iterating forward n times.

3.1 Stable manifold theorem for hyperbolic sets

<u>Theorem 5.2.8</u> There exist collections of C^r manifolds $W^{s,u}_\varphi(x)$ tangent to $E^{s,u}_x$ for all $x \in \Lambda$ such that

a) $y \in W^s_\varphi(x) \implies ||F^n(x) - F^n(y)|| \leq \varphi$

b) $y \in W^u_\varphi(x) \implies ||F^{-n}(x) - F^{-n}(y)|| \leq \varphi$

c) Tangent spaces of $W^{s,u}_\varphi(x)$ are $E^{s,u}_x$

d) There exist constants $c > 0$ and $0 < \lambda < 1$ such that

$$y \in W^s_\varphi(x) \implies ||F^n(y) - F^n(x)|| \leq c\lambda^n$$

where $n \geq 0$

e) $W^{s,u}_\varphi(x)$are embedded disks in $\mathbb{R}^n$ and maps $x \to W^{s,u}_\varphi(x)$ are continuous

[1] Note that F might have to be a diffeomorphism.

4 Lambda lemma

<u>Theorem 5.2.10</u> Λ lemma Suppose f is a C^1 diffeomorphism with a hyperbolic fixed or periodic point p and $W^s(p), W^u(p)$ stable and unstable manifolds of p. Note that $x \in \mathbb{R}^n$ and $\dim W^u(p) = U$.

The lemma states that for a disk D parallel to the unstable manifold and a disk Δ perpendicular to the stable manifold, there exists n such that D and Δ will become arbitrarily close.

Here are some Claude notes since I was confused by Rowley's explanation.

Lambda Lemma Addendum

Setup

At a hyperbolic fixed point p, you have the stable and unstable manifolds $W^s(p)$ and $W^u(p)$. Take:

- A disk Δ that is **transverse to** $W^s(p)$ (i.e. Δ crosses $W^s(p)$), with $\dim \Delta = U = \dim W^u(p)$
- A disk $D \subset W^u(p)$ inside the unstable manifold

What the lemma says

Under repeated forward iteration f^n, the image $f^n(\Delta)$ will C^1-converge to D as $n \to \infty$. That is, $f^n(\Delta)$ becomes **arbitrarily close to a subset of** $W^u(p)$, both in position and in tangent direction.

Why it makes sense

The intuition follows directly from the hyperbolic structure:

- Forward iteration **expands** the unstable direction and **contracts** the stable direction
- So Δ, which has a component pointing in the unstable direction, gets **stretched** along $W^u(p)$ under f^n
- Meanwhile, any component of Δ pointing in the stable direction gets **crushed**
- The result is that $f^n(\Delta)$ flattens out and aligns with $W^u(p)$, converging to D

One subtlety

The lemma requires Δ to be **transverse** to $W^s(p)$, not merely parallel to $W^u(p)$. Transversality ensures Δ has a nonzero component in the unstable direction to begin with, giving forward iteration something to stretch. If Δ were entirely contained in $W^s(p)$, it would instead collapse toward p and never shadow $W^u(p)$.

5 Markov partitions

5.1 Rectangle

Definition: A rectangle R for an invariant set Λ is a closed subset of Λ such that

$$x, y \in R \implies W^s_\varepsilon(x) \Cap W^u_\varepsilon(y)$$

is in Λ and is a unique point.

These rectangles are partitions of Cantor sets, not "whole" rectangles.

5.2 Markov partitions

Definition: A Markov partition for Λ is a finite set of rectangles $\{R_1, \ldots, R_m\}$ such that

- $\Lambda = \bigcup_{i=1}^n R_i$
- $\text{interior}\,(R_i) \bigcap \text{interior}\,(R_j) = \emptyset$ for all $i \neq j$ (disjoint)
- $f\left(W^u\left(x, R_i\right)\right) \supset W^u\left(f(x),\, R_j\right)$ and $f\left(W^s\left(x, R_i\right)\right) \subset W^s\left(f(x),\, R_j\right)$ for all $x \in \text{int}(R_i)$ and $f(x) \in \text{int}(R_j)$

Note that

$$W^u_\varepsilon(x, R_i) = W^u_\varepsilon(x) \bigcap R_i$$

5.3 Theorem 5.3.2

Theorem 5.3.2 Compact maximal, indecomposable hyperbolic invariant sets have Markov partitions.

Indecomposable just means that points can be connected by a chain recurrence, I think.

5.4 Shadowing

5.4.1 Pseudo-orbit

A pseudo-orbit are points x_1, x_2, x_3, x_4 such that

$$||x_{i+1} - f(x_i)|| < \varepsilon \quad \forall i$$

5.4.2 Shadowing

A point y shadows a pseudo orbit $\{x_1, x_2, x_3, x_4\}$ if

$$\left|\left|f^i(y) - x_i\right|\right| < \beta$$

for all i (in some interval).

5.4.3 Proposition 5.3.3

Proposition 5.3.3 For a hyperbolic invariant set, every ε pseudo-orbit is $\beta -$ *shadowed* by a point in Λ.

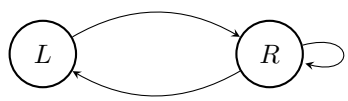
This gives us license to make numerical simulations that will inevitably involve some numerical imprecision.

6 Subshifts of finite type

Essentially, some stings of symbols are not allowed by our dynamics.

6.1 Example of period 3

We've seen this before in the context of the "period 3 implies chaos" example. In that example, we had the following graph



Note that the sting "RRLLR" is not allowed because 'L' cannot transition to itself to make "LL".

Suppose that we have a matrix

$$A_{ij} = \begin{cases} 1 \text{ if can transition from } L \to R \\ 0 \text{ else} \end{cases}$$

then we get a matrix of allowable transitions

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

6.2 Definition

Definition of "subshift of finite type": Let $\{R_1, \ldots, R_n\}$ be a Markov partition for Λ. Define

$$\Sigma_A = \left\{ a \in \{1, 2, \ldots, n\}^{\mathbb{Z}} \text{ such that } A_{a_i\, a_{i+1}} \forall i \right\}$$

Then $\sigma : \Sigma_A \to \Sigma_A$ is called a "subshift of finite type" with transition matrix A.

6.3 Proposition 5.3.4

$$\begin{tikzcd} \Lambda \arrow[r, "f"] \arrow[d, "h"] & \Lambda \arrow[d, "h"] \\ \Sigma_A \arrow[r, "\sigma"] & \Sigma_A \end{tikzcd}$$

Intuition

The big idea is that we are **encoding the dynamics of f on Λ using sequences of symbols**.

The Markov Partition

The Markov partition $\{R_1, \ldots, R_n\}$ chops up the invariant set Λ into n regions. As you iterate a point $x \in \Lambda$ forward under f, it visits these regions in some order, producing a **bi-infinite sequence of symbols**:

$$\ldots R_2, R_1, R_3, R_1, R_2, R_2, \ldots$$

The map $h : \Lambda \to \Sigma_A$ is exactly this encoding — it takes a point and returns the sequence of regions its orbit visits.

The Transition Matrix

Not every sequence of symbols is dynamically realizable. The matrix A encodes which transitions are **geometrically allowed** by the dynamics. In the period-3 example, $L \to L$ never happens because f always maps L outside of itself, so $A_{LL} = 0$.

The Commutative Diagram

The diagram says that h is a **conjugacy** between f and σ, meaning:

$$h \circ f = \sigma \circ h$$

In words: iterating f on Λ and then encoding is the **same as** encoding first and then shifting the sequence left by one. This means:

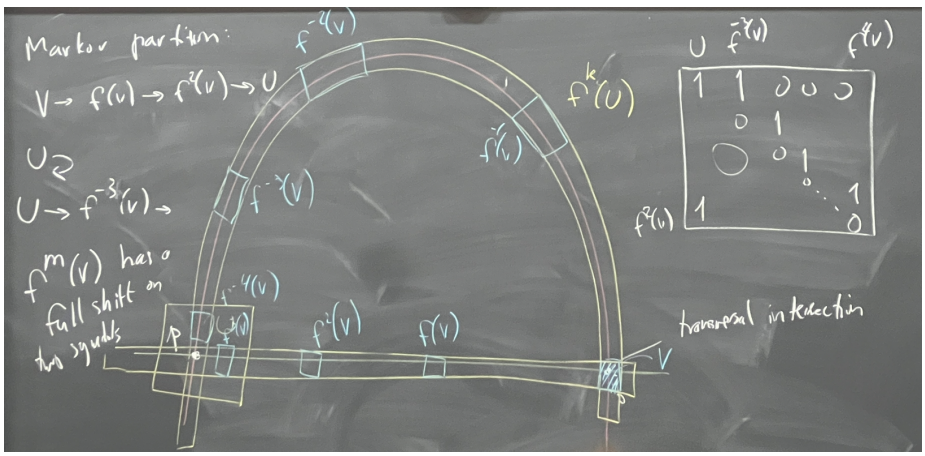
- The dynamics of f on Λ are **completely captured** by σ on Σ_A
- σ is just a left-shift on sequences, which is a much simpler object to study
- Any dynamical property of σ (periodicity, chaos, entropy) **translates directly** back to f

Why This is Powerful

You have replaced a potentially complicated nonlinear map f on a geometric set $\Lambda \subset \mathbb{R}^n$ with a simple **combinatorial shift map** on sequences of symbols. The complexity of the dynamics is now entirely captured by the combinatorics of the transition matrix A.

7 Smale-Birkhoff theorem

Rowley sketched it, and it's so cool that I'm gonna just insert the photo:



The idea is to reason about the transition matrix A about a Markov Partition of $f^m(V) \to U$. For a transversal intersection of the stable and unstable manifolds and some integer m, we have

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

which is conjugate to the shift map, which implies dynamics that are sensitively dependent on initial conditions, and hence, chaos.

Things to tighten up

1. **What $f^m(V) \to U$ means.** The Markov partition is on the two rectangles $\{U, V\}$ near p, not just on $f^m(V) \to U$. The all-ones matrix arises because all transitions $U \to U$, $U \to V$, $V \to U$, and $V \to V$ are allowed.

2. **Why A has all ones.** The transversal homoclinic intersection, combined with the Lambda lemma, is what forces $f^m(V)$ to stretch across **both** U and V, giving all entries equal to 1. The all-ones structure

does not come for free — it is a geometric consequence of the transversal intersection.

3. **Conjugacy to the shift map.** One can be more precise: the dynamics on the invariant set

$$\Lambda = \bigcap_{n \in \mathbb{Z}} f^n(U \cup V)$$

are conjugate to the **full shift on two symbols** $\sigma : \Sigma_2 \to \Sigma_2$, which gives sensitive dependence on initial conditions, a dense set of periodic orbits, and the existence of a dense orbit.